M-PSEUDO-COHERENCE OF A COMPLEX ON A SPACE DESCENDS ALONG FPQC COVERINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12984-12990 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12992-13018, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Object-side (not relative) pseudo-coherence descent for spaces; pairs with the relative-base version collected on the schemes side last wave (L16096) without overlap.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Schemes-side fpqc descent of pseudo-coherence (Derived Categories of Schemes), refinement of fpqc coverings of spaces by schemes (Topologies on Spaces), comparison of D_QCoh on the small etale site with the Zariski derived category and compatibility of pseudo-coherence under it (Derived Categories of Spaces, two lemmas).

Source reference. Original lemma 09UE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13018–13018. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08H4: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13018–13018. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 071Q: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13018–13018. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08HE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13018–13018. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03MS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13018–13018. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0419: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13018–13018. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $\{f_i : X_i \to X\}$ be an fpqc covering of algebraic spaces over S. Let $E \in D_{QCoh}(\mathcal{O}_X)$. Let $m \in \mathbf{Z}$. Then E is m-pseudo-coherent if and only if each Lf_i^*E is m-pseudo-coherent.*

Proof. Pullback always preserves m-pseudo-coherence, see Cohomology on Sites, Lemma 08H4. Thus it suffices to assume Lf_i^*E is m-pseudo-coherent and to prove that E is m-pseudo-coherent. Then first we may assume X_i is a scheme for all i, see Topologies on Spaces, Lemma 0419. Next, choose a surjective étale morphism $U \to X$ where U is a scheme. Then $U_i = U \times_X X_i$ is a scheme and we obtain an fpqc covering $\{U_i \to U\}$ of schemes, see Topologies on Spaces, Lemma 03MS. We know the result is true for $\{U_i \to U\}_{i \in I}$ by the case for schemes, see Derived Categories of Schemes, Lemma 09UE. On the other hand, the restriction $E|_U$ comes from an object of $D_{QCoh}(\mathcal{O}_U)$ (defined using the Zariski topology and the "usual" structure sheaf of U), see Derived Categories of Spaces, Lemma 071Q. The lemma follows as the two notions of pseudo-coherent (étale and Zariski) agree by Derived Categories of Spaces, Lemma 08HE. □